

PHASE 7: PROJECT DESIGN PHASE – PROBLEM SOLUTION FIT & FINAL SOLUTION ARCHITECTURE

Project Details

Particular	Details
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Final Project Design Phase
Date	15 February 2025
Team ID	_____
Prepared By	Koppana Navya

7.1 Problem – Solution Fit Template

Introduction

The Problem–Solution Fit phase identifies the relationship between customer challenges and the proposed analytical solution. The objective of this project is to understand food ordering behaviour, customer preferences, cuisine demand, and delivery trends by transforming raw ordering data into meaningful business insights.

The solution focuses on helping restaurant owners, food delivery platforms, and business analysts make data-driven decisions using interactive Tableau dashboards.

Problem Statement

Food delivery businesses generate large amounts of customer ordering data, but extracting meaningful insights from this data is challenging. Restaurants and delivery platforms need to understand customer preferences, popular cuisines, meal trends, revenue patterns, and delivery performance to improve their services.

Customer Problems Identified

Customer Group	Problem Identified
Restaurant Owners	Difficulty identifying popular cuisines and customer preferences.

Customer Group	Problem Identified
Food Delivery Platforms	Lack of visibility into ordering patterns and delivery efficiency.
Business Analysts	Need for structured analysis of large food ordering datasets.
Decision Makers	Difficulty converting raw data into actionable business strategies.

Proposed Solution Fit

Problem	Solution
Lack of understanding of customer food preferences	Cuisine and customer behaviour analysis using Tableau visualizations.
Difficulty tracking order patterns	Interactive dashboards showing meal type and company trends.
Limited understanding of revenue contribution	Revenue analysis through charts and KPIs.
Difficulty evaluating delivery performance	Delivery time analysis visualization.
Complex customer segmentation	Age Group calculated field for demographic analysis.

Benefits of Problem-Solution Fit

- Helps businesses understand customer behaviour.
- Enables data-driven decision-making.
- Improves customer satisfaction through better service planning.
- Identifies high-performing cuisines and meal categories.
- Supports strategic business growth.

7.2 Proposed Solution Template

S.No	Parameter	Description
1	Problem Statement	Food ordering platforms lack proper analytical methods to understand customer behaviour, cuisine demand, revenue patterns, and delivery efficiency.
2	Idea / Solution Description	Develop an interactive Tableau-based analytics solution that processes food ordering data and provides dashboards, KPIs, charts, and business insights.
3	Novelty / Uniqueness	The solution combines customer segmentation, cuisine analysis, revenue insights, delivery analysis, and interactive filtering in a single dashboard platform.
4	Social Impact / Customer Satisfaction	Enables restaurants and delivery platforms to improve customer experience by understanding preferences and optimizing services.
5	Business Model (Revenue Model)	The solution can support subscription-based analytics services for restaurants, food delivery companies, and business intelligence platforms.
6	Scalability of Solution	The system can handle larger datasets, additional locations, more customer attributes, and advanced predictive analytics in future versions.

7.3 Solution Architecture

Introduction

The Solution Architecture defines the technical structure of the Food Ordering Behaviour analytics system. It describes how data moves from collection to processing, visualization, and user access.

The architecture combines data analytics, visualization, and web integration technologies to deliver an interactive decision-support system.

Solution Architecture Diagram

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| Food Ordering Data |

| CSV Dataset |

| (50,000 Records) |

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| Data Preparation |

| Cleaning & |

| Transformation |

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| Tableau Desktop |

| Data Analysis & |

| Visualization |

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| Tableau Dashboard & Story |

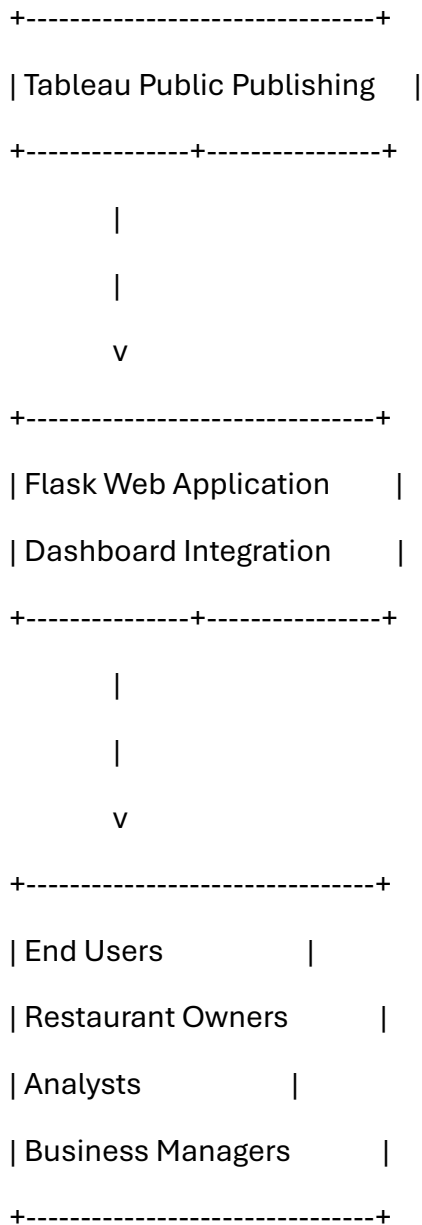
| KPI | Charts | Filters |

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Architecture Components

Component	Description	Technology
Data Source	Stores food ordering information	CSV Dataset
Data Processing	Cleans and prepares data	Tableau Data Preparation
Analytics Layer	Performs analysis and calculations	Tableau Desktop
Visualization Layer	Creates charts, KPIs, dashboards	Tableau
Publishing Layer	Hosts dashboard online	Tableau Public

Component	Description	Technology
Web Layer	Provides browser-based access	Flask Framework
User Interface	Displays dashboard content	HTML, CSS
Development Language	Backend integration	Python

Data Flow Description

Step 1: Data Collection

The food ordering dataset is collected from a reliable data repository in CSV format.

Step 2: Data Preparation

The dataset is cleaned by handling missing values, standardizing fields, and creating calculated attributes such as Age Groups.

Step 3: Data Analysis

The prepared dataset is analyzed using Tableau Desktop to identify:

- Cuisine demand.
- Customer preferences.
- Order distribution.
- Revenue contribution.
- Delivery trends.

Step 4: Dashboard Creation

Multiple visualizations are combined into an interactive dashboard containing KPIs, charts, filters, and analytical views.

Step 5: Deployment

The completed Tableau dashboard is published to Tableau Public and integrated into a Flask web application.

Step 6: User Access

End users access the dashboard through a browser to explore insights and support business decisions.

Final Solution Features

Analytical Features

- Customer behaviour analysis.
- Cuisine preference analysis.
- City-wise order analysis.
- Meal type performance analysis.
- Revenue contribution analysis.
- Delivery efficiency analysis.

Dashboard Features

- Interactive filters.
- KPI cards.
- Multiple chart types.
- Story-based insights.
- User-friendly interface.

Conclusion

The Final Project Design Phase establishes the complete problem-solution relationship and technical architecture of the Food Ordering Behaviour and Consumer Trends project. The proposed Tableau analytics solution addresses the challenge of extracting meaningful insights from large food ordering datasets. Through structured data processing, interactive dashboards, and web accessibility, the solution supports restaurants, delivery platforms, and decision-makers in improving operational efficiency and customer satisfaction.

The architecture provides a scalable foundation that can be extended with additional datasets, predictive models, and advanced analytics capabilities in future developments.